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From prediction to intervention: causal digital twins for personalized clinical decision support

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Abstract

Objective This work proposes a unified framework for causal digital twins, integrating Structural Causal Models (SCMs), the Potential Outcomes Framework, and reinforcement learning. The objective is to enable individualized counterfactual reasoning and dynamic treatment policy optimization, moving beyond prediction toward actionable, ethical, and adaptive decision support.

Materials and methods This work formalizes causal digital twins by embedding causal inference principles into digital architectures, modeling both observed and counterfactual outcomes for each patient. Individualized Treatment Effects (ITE) are estimated through causal modeling, and sequential decision-making is optimized using reinforcement learning techniques. This work is a methodological and conceptual research contribution. We propose a unified framework for the construction and evaluation of clinical digital twins and a practical checklist covering validation, safety, and ethical requirements. No new dataset or algorithmic benchmark is introduced.

Results This work discusses methodological requirements, learning objectives, evaluation strategies, and ethical considerations necessary for deploying causal digital twins in real-world clinical contexts. The proposed framework allows for accurate estimation of counterfactual outcomes, personalized treatment effect modeling, and dynamic policy learning over time.

Discussion Through this integration, digital twins evolve from static predictors into active engines of individualized intervention, providing a new paradigm for precision healthcare. Causal digital twins offer a transformative extension of current digital twin technologies, merging causal reasoning with dynamic simulation to enable personalized, counterfactual-driven clinical decision support.

Conclusion This work lays the methodological foundation for operationalizing causal digital twins in future AI-driven healthcare systems, with profound implications for ethical, individualized, and adaptive medicine.

Background and significance

The advent of digital twins (DTs) in healthcare represents a milestone in the convergence of biomedical data science and individualized medicine. Traditionally rooted in engineering and industrial applications, DTs have been adapted to simulate the physiological states and disease trajectories of patients, leveraging multimodal

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data from electronic health records, imaging, genomics, and wearable sensors. These models have demonstrated substantial utility in predictive analytics, particularly in forecasting clinical events and optimizing resource allocation. However, such approaches predominantly rely on correlational inference and often lack the capacity to answer counterfactual clinical questions, those that underlie treatment selection, risk-benefit evaluation, and ethical justification. In parallel, causal inference has evolved into a robust mathematical discipline for estimating treatment effects and simulating hypothetical interventions, through frameworks such as Structural Causal Models (SCMs) and the Potential Outcomes Framework. These tools allow for principled reasoning about "what if" scenarios, essential in medical decision-making but largely absent from conventional DT implementations. The integration of causal inference into digital twin architectures gives rise to a new paradigm: causal digital twins. Unlike their predictive counterparts, causal DTs simulate not only the likely progression of a patient's condition but also the expected outcomes under alternative interventions. This capability transforms DTs from passive forecasting tools into active engines of clinical reasoning, capable of supporting individualized treatment policies and adaptive therapeutic strategies. This conceptual shift is particularly significant in precision medicine, where heterogeneity in treatment response, ethical imperatives of transparency, and the demand for explainable AI are paramount. Causal digital twins promise to unify interpretability, personalization, and dynamic policy learning within a single computational framework, thereby setting the stage for next-generation decision support systems in healthcare.

Keywords Causal inference, Digital twins, Counterfactual reasoning, Structural causal models (SCMs), Potential outcomes framework, Individualized treatment effect (ITE), Reinforcement learning, Precision medicine, AI-driven healthcare, Personalized decision support

Introduction

In the evolving landscape of precision medicine, digital twins (DTs) have emerged as powerful tools for modeling, simulating, and forecasting individual health trajectories [1, 2]. Traditionally, DTs are deployed to replicate the physiological and pathological states of patients by leveraging high-dimensional data from clinical, biological, and behavioral sources [3–5]. These models are primarily focused on *descriptive* and *predictive* analytics, where machine learning algorithms are trained to anticipate future health outcomes based on historical data [6, 7]. While such approaches are undeniably valuable, they often fall short in addressing a fundamental question at the heart of clinical decision-making: "*What would have happened if a different intervention had been chosen?*".

This question pertains to the realm of *causal inference*, which distinguishes correlation from causation and allows the estimation of *counterfactual outcomes* [8, 9]. In medical practice, treatment decisions hinge on counterfactual reasoning, comparing the observed outcome of a patient who received treatment "A" with the unobserved, hypothetical outcome had they received treatment "B" [10]. Traditional DTs, despite their predictive prowess, lack the theoretical structure to answer such questions reliably [11].

This gap motivates the integration of formal causal frameworks into the digital twin paradigm. Specifically, this work proposes the development of *causally informed digital twins* by embedding (1) structural causal models (SCMs), as introduced by Judea Pearl [8], which use directed acyclic graphs (DAGs) and structural equations to represent and manipulate causal relationships, (2) and the potential outcomes framework, or Rubin Causal Model [9, 12], which defines treatment effects as comparisons between potential outcomes under different treatment conditions.

By enriching DTs with causal inference capabilities, we can transition from "what is likely to happen?" to "what would happen if.?", a shift that is essential for personalized, ethically grounded, and evidence-based medical decision-making [13, 14].

This article aims to establish a methodological and conceptual foundation for causal digital twins. This article argues that these hybrid models are not only technically feasible but necessary for the future of AI-driven healthcare. In the following sections, this article explores the theoretical frameworks that underpin causal inference, describes how these can be operationalized within digital twin architectures, and presents methodologies for training, evaluating, and deploying causally-enabled DTs in real-world clinical contexts.

Although this manuscript follows the structure of a research article, its contribution is methodological rather than empirical. The objective is not to introduce a new dataset, develop a novel machine-learning architecture, or report benchmark performance.

Instead, this article aims to:

(i) Formalize an integrative conceptual framework for clinical digital twins,
(ii) Synthesize current methodological requirements across modeling, data integration, and validation, and.
(iii) Propose an operational evaluation and ethical checklist to support clinical deployment and regulatory assessment.

This contribution addresses the current fragmentation of definitions and evaluation practices in the field and provides a reusable reference for researchers, clinicians, and health authorities.

Statement of Significance:

Problem or issue Traditional digital twins in healthcare are predominantly predictive and do not support causal reasoning or individualized counterfactual simulation, limiting their utility for clinical decision-making.

What is already known Structural causal models (SCMs) and the potential outcomes framework allow rigorous estimation of treatment effects and counterfactuals. However, they have not been systematically integrated into digital twin architectures.

What this paper adds This paper proposes a unified framework for *causal digital twins*, combining SCMs, potential outcomes, and reinforcement learning to enable dynamic, individualized treatment optimization. It defines methodological pipelines and evaluation metrics for real-world deployment.

Who would benefit Researchers and developers in biomedical informatics, precision medicine, and AI-driven healthcare systems seeking to build interpretable, adaptive, and ethically sound clinical decision-support tools.

Before detailing the formal framework, we present a short clinical vignette to illustrate how causal estimands (ITE/CATE), counterfactual reasoning, and digital twin simulations are operationalized in practice (Table 1).

Theoretical framework

In contrast to purely predictive models, which estimate what is likely to happen given observed data, causal inference aims to answer intervention-based questions such as: "*What would happen to this patient if we actively changed the treatment?*" [15].

For example, a predictive digital twin may learn that patients receiving treatment A have worse outcomes than those receiving treatment B. However, this association alone does not imply that treatment A is harmful, as patients receiving A may be systematically sicker. Causal inference explicitly models such confounding mechanisms and seeks to estimate outcomes under hypothetical interventions, answering questions such as "*What would the outcome have been for this same patient had treatment B been given instead?*".

Structural causal models formalize this reasoning by separating observed associations from causal effects, enabling digital twins to simulate counterfactual outcomes that are essential for individualized treatment decisions [10, 14].

Structural Causal Models are mathematical frameworks used to model and analyze causal relationships between variables. They consist of structural equations that express each variable as a deterministic function of its direct causes (or parents) and a stochastic noise term, capturing unobserved factors (Fig. 1).

Let:
$\mathcal{U}$: set of exogenous (external) variables, representing unobserved background factors,
$\mathcal{V} = \{X, T, Y\}$set of endogenous (observed) variables where.
X: observed covariates (e.g., age, biomarkers),
T: treatment indicator (e.g., medication administered),
Y: outcome of interest (e.g., survival, symptom improvement).

Each endogenous variable $\mathcal{V}_i \in \mathcal{V}$ is defined by a structural equation:

$$V_i = f_i\left(Pa(V_i),\, U_i\right)$$

Where:
$Pa\left(V_i\right)$ are the parents of V_i in the causal graph,
U_i is an exogenous function specifying how the value of V_i depends on its parents and exogenous noise.

These dependencies can be represented as a Directed Acyclic Graph (DAG), where arrows represent causal influences.

A DAG is a graphical representation used in causal inference to encode assumptions about the causal relationships between variables.

- Directed means each edge (arrow) goes from one variable to another, indicating the direction of causality (e.g., A→B means A causes B).
- Acyclic means there are no cycles, that is, you cannot start at one node and follow a series of arrows that eventually loops back to the starting node.

Each node in a DAG represents a variable (e.g., treatment, outcome, covariate), and each arrow represents a direct causal influence. DAGs are useful for:

- Identifying confounders,
- Applying rules like the back-door and front-door criteria,
- Visualizing and validating causal assumptions.

DAGs form the backbone of Structural Causal Models (SCMs) as introduced by Judea Pearl.

Table 1 Clinically grounded worked example of a causal digital twin for individualized treatment decisions

Clinical scenario

A 62-year-old woman is admitted for ST-segment elevation myocardial infarction. After successful revascularization, her left ventricular ejection fraction is 35%, systolic blood pressure is 102 mmHg, estimated glomerular filtration rate is 58 mL/min/1.73 m^2, and NT-proBNP is elevated. Current guidelines recommend early initiation of beta-blocker therapy, but the optimal dose is uncertain because of borderline hypotension and renal vulnerability. The clinician must choose between:

- **Treatment A**: high-dose beta-blocker initiation within 24 h
- **Treatment B**: standard-dose beta-blocker initiation

Target causal estimand

The digital twin estimates the individual treatment effect (ITE) on a clinically relevant outcome:

$ITE_i = \mathbb{E}[Y_i(30d \mid A)] - \mathbb{E}[Y_i(30d \mid B)]$

where $Y(30d)$ denotes the probability of death or heart-failure hospitalization within 30 days.

For population stratification or guideline refinement, the system may additionally estimate the conditional average treatment effect (CATE):

$CATE(x) = \mathbb{E}[Y(30d \mid A, X = x)] - \mathbb{E}[Y(30d \mid B, X = x)]$

for patients with similar covariate profiles X (age, LVEF, blood pressure, renal function, biomarkers).

Counterfactual question addressed

For this specific patient, what would her 30-day outcome be if she were treated with high-dose beta-blockers compared with standard-dose therapy?

Digital twin instantiation and simulation

1. State initialization:
   The patient's clinical, biological, and imaging variables are mapped to the twin's state vector.
2. Model components:
   - Short-term cardiovascular dynamics (heart rate, blood pressure, ventricular remodeling)
   - Drug–response submodel (dose–response relationship)
   - Statistical risk model for 30-day adverse events
3. Intervention simulation:
   The twin is simulated twice under Treatment A and Treatment B, generating two potential outcome trajectories.

Key causal and modeling assumptions

1. Consistency: the modeled interventions correspond to real clinical beta-blocker regimens.
2. Conditional exchangeability: given observed covariates, treatment assignment is independent of potential outcomes.
3. Positivity: similar patients exist in historical data for both dose strategies.
4. Correct model specification: physiological and statistical submodels adequately approximate short-term cardiovascular responses.
5. Temporal stability: treatment effect mechanisms remain stable over the modeled horizon.

Illustrative result (hypothetical)

The digital twin estimates:

- $P(Y = 1 \mid A) = 8.3\%$
- $P(Y = 1 \mid B) = 12.5\%$

Thus:

$ITE_i = -4.2\% \text{ (absolute risk reduction)}$

Predicted risk of severe hypotension remains below 3% under Treatment A.

Impact on the clinical decision pathway

Step	Standard care	Digital-twin–augmented care
1	Guideline recommendation	Guideline recommendation
2	Clinical judgment	Twin calibration
3	Fixed dose selection	Simulation of A vs. B
4	—	ITE + uncertainty estimate
5	Treatment decision	Personalized treatment decision

The digital twin transforms a population-level recommendation into an individualized, causally interpretable risk–benefit estimate, supporting more precise treatment selection.

Translational relevance

This example illustrates how digital twins operationalize causal inference at the bedside by:

- Making counterfactual outcomes explicit,
- Quantifying individualized benefit and harm, and
- Providing transparent assumptions for clinical and regulatory evaluation.

Intuitively, the do-operator represents an external intervention that forces a treatment choice, mimicking an idealized randomized experiment and breaking all natural or confounded associations that normally influence treatment assignment.

A SCM is a formal framework used to represent and reason about cause-and-effect relationships between variables.

A SCM consists of three components:

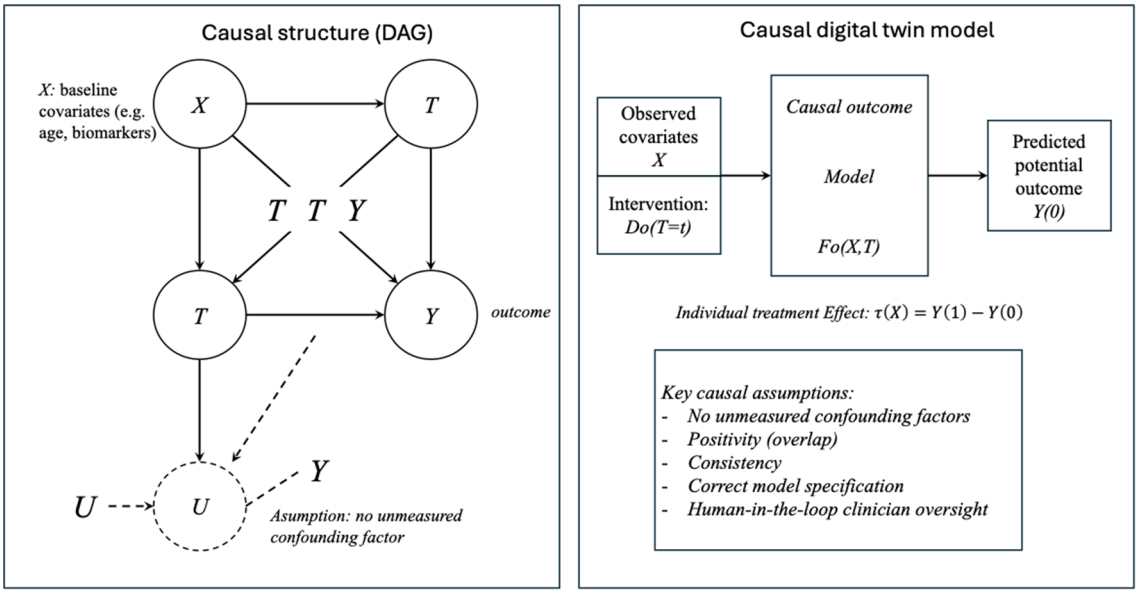

Fig. 1 Explicit representation of the causal structure, model parameters, and core assumptions underlying a causal digital twin. The directed acyclic graph (DAG) depicts the assumed causal relationships between baseline covariates X, treatment assignment T, and outcome Y. Unobserved or latent factors U, which may influence both treatment assignment and outcomes, are represented using dashed elements, illustrating the potential source of confounding and the corresponding exchangeability (no unmeasured confounding) assumption. The computational component represents the causal outcome model fθ(X, T), which estimates potential outcomes Y(t) under hypothetical interventions do(T = t), with t ∈ {0,1}. The individual treatment effect (ITE) for a patient with covariates X is defined as τ(X) = Y(1) − Y(0), while the conditional average treatment effect (CATE) corresponds to E[Y(1) − Y(0) | X]. Key assumptions required for valid counterfactual inference are explicitly indicated, including consistency (alignment between modeled and real-world interventions), positivity (non-zero probability of receiving each treatment across covariate strata), and correct model specification. The figure illustrates how causal structure, modeling assumptions, and parameterization jointly determine the identifiability, reliability, and clinical interpretability of counterfactual predictions generated by the digital twin

1. *A set of variables*:

Endogenous variables: variables of interest within the system (e.g., treatment T, outcome Y).

Exogenous variables: external, unobserved factors (U) that influence endogenous variables.

2. *Structural equations*:

Each endogenous variable is defined by a deterministic or stochastic function of its direct causes and an exogenous noise term.
For example:

$$T = f_t(X, U_T), \quad Y = f_Y(T, X, U_Y)$$

where X are covariates, and U_T, U_Y are exogenous variables.

3. *A causal graph* (typically a DAG) representing the functional dependencies among variables.

SCMs allow us to:

– Simulate interventions using the do-operator $\mathrm{do(T = t)}$.
– Derive counterfactuals (e.g., $Y_{T=0}(U)$),
– Apply identification criteria like back-door and front-door adjustments.

SCMs provide a mathematically rigorous language for distinguishing association from causation, and are foundational in modern causal inference, particularly as formulated by Judea Pearl.

A key tool within SCMs is the do-operator, $do(T = t)$, which represents an intervention that forcibly sets *T* to *t*, thereby breaking its natural dependencies. The do-operator, introduced by Judea Pearl, is a formal notation used in causal inference to represent interventions in a system.

This expression means that the variable T is forcibly set to the value ttt, as if by intervention, independent of its natural causes.

This allows the estimation of causal effects:

$$\mathrm{P(Y \mid \ do(T = t) \neq P(Y \mid \ T = t)}$$

This inequality illustrates that observing someone who received $\mathrm{T = t}$ does not reveal the causal effect unless we intervene. The causal effect of T on Y is best described using the Average Treatment Effect (ATE):

$$ATE = \mathbb{E}\left[Y \mid \ \mathrm{do}(\mathrm{T} = \mathrm{t})\right] - \mathbb{E}\left[Y \mid \ \mathrm{do}(\mathrm{T} = 0)\right]$$

The notation $\mathrm{do}(\mathrm{T} = \mathrm{t})$ indicates an intervention, breaking all natural (possibly confounded) causes of T.

This represents the average expected change in the outcome Y if the entire population were to receive treatment compared to no treatment. The do-notation emphasizes that these expectations are based on intervention, not just statistical associations.

This equation captures the expected difference in outcomes if everyone in the population were to receive treatment T = 1 versus if they received T = 0.

The ATE measures the causal effect of the treatment across a population. If: ATE > 0, then the treatment has a positive effect on the outcome on average. If: ATE < 0, then the treatment has a negative (possibly harmful) effect on average.

The notation $\mathrm{do}(\mathrm{T} = 1)$ introduced by Judea Pearl, denotes an intervention rather than passive observation. Thus, $\mathbb{E}\left[Y \mid \ \mathrm{do}(\mathrm{T} = \mathrm{t})\right]$ expresses the mean outcome when the treatment is forcibly set to ttt, breaking any natural associations between T and other variables.

The ATE quantifies the causal effect of the treatment and assumes a valid underlying causal model in which confounding variables are accounted for, typically through methods like:

- back-door adjustment using DAGs,
- propensity score weighting or matching,
- or the use of instrumental variables.

This measure is fundamental in evaluating treatment efficacy and supports simulation of counterfactuals within a digital twin framework, allowing personalized comparisons between therapeutic strategies.

To compute ATE, one must often adjust for confounding variables, those affecting both the treatment and the outcome. SCMs help identify appropriate adjustments (e.g., via the back-dorr criterion) to ensure unbiased causal estimation.

To estimate ATE from observational data, one must adjust for confounders, variables that influence both the treatment T and the outcome Y. The back-door criterion, introduced by Judea Pearl, provides a graphical method to determine which variables should be adjusted for.

To estimate causal effects from observational data, confounding must be addressed. A set satisfies the back-door criterion if:

1. No element of is a descendant of T,
2. Z blocks all back-door paths from T to Y.

This enables adjustments:

$$P(Y \mid \ \mathrm{do}(\mathrm{T})) = \sum{}_{z} P(Y \mid \ \mathrm{T}, \mathrm{Z} = \mathrm{z})\mathrm{P}(\mathrm{Z} = \mathrm{z})$$

Where:
do(T) is the do-operator indicates an intervention in the system where the treatment T is forcibly set, breaking any natural causal influences into T.
Z is a set of covariates (confounders) that block all back-door paths from T to Y, meaning they prevent non-causal associations from biasing the estimated effect.
$P(Y \mid \ \mathrm{T}, \mathrm{Z} = \mathrm{z})$ is the conditional probability of the outcome Y given that treatment T and confounders Z take specific values. This is something we can estimate directly from the observed data.
$\mathrm{P}(\mathrm{Z} = \mathrm{z})$ is the marginal probability of the confounder Z taking the value z.
$\sum{}_{z}$. is a summation over all values of Z (or an integral if Z is continuous), which combines the conditional probabilities into an overall adjusted effect.

Example DAG:

$$X \rightarrow T \rightarrow Y$$

$$X \rightarrow Y$$

Here, X is a common cause of both T and Y, opening the back-door path $T \leftarrow X \rightarrow Y$. To estimate the true causal effect of T on Y, we must adjust for X, which blocks this confounding path.

Simulated example of ATE calculation

Suppose a synthetic data generation process:

$$X \sim \mathcal{N}(0,1)$$

$T \sim Bernouilli\,(\sigma\,(0.5X))$, where σ is the logistic function

$$Y = 2T + X + \varepsilon\,,\ with\,\varepsilon \sim \mathcal{N}(0,1)$$

The true causal model implies:

$$\mathbb{E}\left[Y \mid \ \mathrm{do}(\mathrm{T} = 1)\right] - \mathbb{E}\left[Y \mid \ \mathrm{do}(\mathrm{T} = 0)\right] = 2$$

A naïve model without adjustment would confound this with the effect of X. A regression adjusting for X, such as:

$$Y \sim \beta_{o} + \beta_{1}T + \beta_{2}X + \epsilon$$

Recovers $\beta_2 \approx 2$, the true ATE. This shows that adjusting for the appropriate back-door variable (X) yields an unbiased estimate of the treatment effect.

Advanced concepts un SCMs

Alternative identification criteria.

Front-door criterion: if a mediator M exists such that $T \rightarrow M \rightarrow Y$.

And M blocks all back-door paths, then:

$$P(Y \mid \text{do}(\text{T})) = \sum_m P(M \mid \text{T}) \sum_t P(Y \mid \text{M}, \text{T} = \text{t})\text{P}(\text{T} = \text{t})$$

Where:
T is the treatment or intervention variable.
Y is the outcome variable of interest.
M is a mediator, a variable through which the treatment T influences the outcome Y.
$\text{do}(\text{T})$ is the do-operator indicating an idealized intervention where T is set externally.
$P(M \mid \text{T})$ is the probability of the mediator M given the treatment T. This captures how treatment affects the mediator.
$P(Y \mid \text{M}, \text{T} = \text{t})$ is the conditional probability of the outcome given both the mediator M and the treatment value t.
$\text{P}(\text{T} = \text{t})$ is the marginal (observed) probability distribution of treatment assignment in the population.
Σm, Σt are summations over all possible values of M and T, respectively.

Instrumental variables (IV): a variable Z that affects T but not Y directly (only through T) allows identificatiion via:

$$Causal\ effect = \frac{Cov(Y, Z)}{Cov(T, Z)}$$

d-separation: this graphical criterion determines wether two variables are conditionnaly independent given a set Z in a DAG.

Key rules:

- Chains $A \rightarrow B \rightarrow C$ or forks $A \leftarrow B \rightarrow C$ are blocked if $B \in Z$.
- Colliders $A \rightarrow B \leftarrow$ C are blocked unless $B \in Z$ or one of its descendants is.

In digital twin applications, SCMs allow the simulation of alternate treatment scenarios by intervening on variables and observing the resulting changes across the modeled system, creating a robust framework for generating and comparing counterfactual outcomes.

Structural counterfactuals

A SCM can also express counterfactuals, e.g., what would Y have been if T = t'?

This as constructed as:

$$Y_{T=t'}(U) = f_Y(f_T(X, U_T), U_Y)$$

Where:
$Y_{T=t'}(U)$ is the counterfactual outcome, what the value of Y would have been if the treatment T had been set to t′, given a fixed background U.
$f_Y(f_T(X, U_T)$ is the structural function defining how the treatment T is determined based on observed covariates X and exogenous variables U_T.
$f_Y(, U_Y)$ is the structural function defining the outcome Y based on its inputs (possibly including treatment and other covariates) and its own exogenous noise term U_T.
U is the collection of all exogenous (unobserved) variables in the system, typically fixed per individual in counterfactual reasoning.

This capacity to simulate alternate realities is fundamental to the concept of digital twins used for causal inference.

Potential outcomes framework

The Potential Outcomes Framework, also known as the Rubin Causal Model, is a cornerstone of modern causal inference. It frames the causal effect of a treatment as a comparison between the potential outcomes that would occur under different treatment assignments.

Definition of potential outcomes

For each individual, define two potential outcomes:

$Y_i(1)$: the outcome if the individual receives te treatement.
$Y_i(0)$: the outcome if the individual does not receive the treatment.

However, we can only observe one of these outcomes for each person, either $Y_i(1)$ or $Y_i(0)$, depending on the actual treatment assignment Y_i. This is known as the "fundamental problem of causal inference".

Causal estimands

Individual treatment effect (ITE):

$$ITE_i = Y_i(1) - Y_i(0)$$

Where:
$Y_i(1)$ is the potential outcome for individual i if treated.
$Y_i(0)$ is the potential outcome for individual i if not treated.

ITE_i is the difference in outcomes for that individual, representing the personal causal effect of the treatment.

The ITE quantifies the causal effect of a treatment on a specific individual. It is defined as the difference between two potential outcomes for the same person: one under treatment and one under no treatment.

And for interpretation:
ITE_i >0: the individual benefits from the treatment.
ITE_i <0: the treatment may be harmful or less effective than no treatment.
ITE_i=0: the treatment has no effect for this person.

Average Treatment Effect (ATE):

$$ATE = \mathbb{E}\left[Y\left(1\right)\right] - \left[Y\left(0\right)\right]$$

Where:
$Y\left(1\right)$is the potential outcome if treated.
$Y\left(0\right)$ is the potential outcome if not treated.
E[·] is the expectation or average over the population.

For interpretation:
ATE captures the causal effect of a treatment on average across the population.
It is useful for policy decisions and public health planning.

For example:
If ATE: the treatment has a positive average effect.
If ATE < 0: the treatment may be harmful on average.

Thus, the ATE measures the expected difference in outcomes between a population that receives a treatment and the same population if it had not received the treatment.

Conditional Average Treatment Effect (CATE):

$$CATE\left(x\right) = \mathbb{E}\left[Y\left(1\right) - Y\left(0\right) \mid \ \mathrm{X} = \mathrm{x}\right]$$

Where:
$Y\left(1\right)$ is the potential outcome under treatment.
$Y\left(0\right)$ is the potential outcome under no treatment.
X is a vector of covariates (e.g., age, sex, disease severity).
E[·|X = x] is the expectation conditioned on individuals with characteristics x.

The CATE measures the expected treatment effect for individuals with covariate profile $\mathrm{X} = \mathrm{x}$, enabling precision medicine applications.

CATE quantifies how the treatment effect varies across subgroups defined by covariates: useful for precision medicine and targeted interventions. For example, the treatment may benefit younger patients more than older ones, or only those with a specific biomarker.

Key assumptions for identification

To estimate these quantities from observational data, the following assumptions are required:

Ignorability (Unconfoundedness): (Y(1), Y(0)) $\perp$ T | X.

This implies that, conditional on covariates X, the treatment assignment T is a good as random.
Positively (Overlap): $0 < P(T = 1 \mid \ \mathrm{X} = \mathrm{x}) < 1$ for all x. This ensures that each individual has a non-zero probability of receiving either treatment.
Consistency: $Y = Y\left(1\right)T \ + \ Y\left(0\right)\left(1 - T\right)$ This links the observed outcome Y to the potential outcomes, stating that the observed outcome corresponds to the potential outcome under the treatment actually received.

Estimation methods

Various statistical methods can be used to estimate ATE and CATE:

- Regression adjustment: Fit models separately for Y(1) and Y(0);
- Propensity score matching: Match treated and untreated individuals with similar covariates;
- Inverse probability weighting:

$$A\widehat{T}E = \frac{1}{n}\sum\nolimits_{i=n}^{n}\left(\frac{T_iY_i}{e\left(X_i\right)} - \frac{\left(1 - T_i\right)Y_i}{1 - e\left(X_i\right)}\right)$$

where $e\left(X_i\right) = P(T_i = 1 \mid \ X_i)$ is the estimated propensity score.

- Doubly Robust Estimators: Combine regression and IPW for more robust results.
- Machine Learning Models: Use flexible models (e.g., random forests, causal forests, neural nets) to estimate Y(1) and Y(0) and compute ITE.

This framework enables the estimation of treatment effects at both the population and individual levels, forming the basis for counterfactual reasoning in digital twins.

Worked example, consider a toy dataset:

ID	X (covariate)	T (treatment)	Y (outcome)
1	0	1	5
2	0	0	3
3	1	1	7
4	1	0	4

$$CATE(X = 0) = 5 - 3 = 2$$

$$CATE\left(X = 1\right) = 7 - 4 = 3$$

$$ATE = (2+3)/2 = 2.5$$

Advanced estimators and meta-learners.
S-learner: Fit a single model for $Y \sim f(X,T)$.
T-learner: Separate models for Y(1) and Y(0).
X-learner: combines outcome models and impute treatment effects.
R-learner: minimizes a pseudo-outcome loss function.
Causal forests: heterogeneous effect estimation with tree-based model.
Bayesian additive regression trees (BART): probabilistic inference to ITEs.

ITEs, or Individual Treatment Effects, represent the difference between the two potential outcomes for a single individual under two treatment conditions:

$$ITE_i = Y_i(1) - Y_i(0)$$

Where:
$Y_i(1)$ is the outcome of the individual were treated.
$Y_i(0)$ is the outcome if the individual were not treated.

ITE is a personalized measure of how an intervention affects a specific person. However, in practice, we can only observe one of these outcomes for any given individual, which is the essence of the fundamental problem of causal inference, we never observe both outcomes simultaneously for the same person.

Benchmarked datasets:

- IHDP (Infant Health and Development Program).
- Twins (US twin birth outcomes).
- Jobs dataset (LaLonde).

Visual models: SWIGs

Single World Intervention Graphs (SWIGs) integrate DAGs with potential outcomes, making assumptions like ignorability and consistency explicit in graphical form. They clarify which variables are counterfactual, observed, or part of the intervention.

Connection to policy learning

Potential outcomes form the basis for learning individualized treatment rules.
A decision rule $d(X)$ can be evaluated by:

$$Value(d) = \mathbb{E}[Y(d(X))]$$

This is closely linked to reinforcement learning and policy optimization in digital twins.

Limitations:

- Ignorability is untestable without experimental data.
- Potential outcomes do not express mediators or mechanisms.
- Does not naturally extend to longitudinal or time-varying treatments.

Digital twins as causal digital twins

Digital twins, when enriched with causal inference frameworks, become powerful tools not only for prediction but also for simulating alternative realities — that is, counterfactuals. In this role, digital twins act as "causal digital twins": computational systems capable of estimating both observed and unobserved potential outcomes for an individual, thereby quantifying the effects of interventions.

Formal definition

Let $\mathcal{D}_i$ denote the digital twin of individual i.

$\mathcal{D}_i$ of an individual i as a computational proxy capable of simulating counterfactuals:

$$\mathcal{D}_i = \{Y_i(t) \ : \ t \in \mathcal{T}\}$$

Where $\mathcal{T}$ is the set of possible treatments (e.g., $\{0,1\}$

The twin maps an individual's covariate X_i and a potential treatment value $t \in \{0,1\}$ to an outcome:

$$\widehat{Y}_i(t) : \mathcal{D}_i = \widehat{f}(X_i, t) \approx \mathbb{E}[Y_i(t) \mid X_i]$$

Where:
$\widehat{Y}_i(t)$ is the estimated potential outcome for individual iii under treatment t $\in \{0,1\}$, this could be either the factual or counterfactual outcome.

$\mathcal{D}_i$ is the digital twin for individual i, representing a computational model that simulates their behavior or response under different treatment conditions.
$\widehat{f}(X_i, t)$ is a learned function (e.g., a machine learning model) that takes covariates $X_i$and treatment value t, and returns a predicted outcome. It approximates the true conditional expectation.

$\mathbb{E}[Y_i(t) \mid X_i]$ is the CATE at X_i; the expected value of outcome Y for treatment t, conditioned on covariates X_i.

This model emulates both factual and counterfactual outcomes, aligned with the CATE principle.

Learning objectives

Digital twin models are trained using observational or experimental data (X_i, T_i, Y_i). Since only the factual

outcome $Y_i(T_i)$ is observed for each individual, the challenge lies in learning a model $\widehat{f}$ that generalizes to counterfactuals. Learning objectives include:

- Accurate estimation of potential outcomes,
- Minimzing counterfactual prediction error,
- Avoiding covariate imbalance between treatment groups.

Causal model architectures

Several machine learning architectures are used for causal emulation:

- TARNet: shared representation with separate heads for Y(0) and Y(1).
- Dragonnet: adds propensity score prediction to enforce balancing.
- Causal forests: trees ensemble for estimating heterogeneous treatment effects.
- GANITE: a generative adversarial model for imputing counterfactuals.
- BART and X-learners: for flexible nonparametric causal modeling.

Counterfactual queries and emulation

Once trained, a digita twin $\widehat{f}(X_i, t)$ can be queried for any treatment value:

$$ITE_i = \widehat{f}(X_i, t) - \widehat{f}(X_i, 0)$$

Where:
$\widehat{f}(X_i, t)$ is the predicted outcome from the trained digital twin model when individual i, with covariates X_i, is assigned treatment ttt (e.g., t = 1 for treated).
$\widehat{f}(X_i, 0)$ is the predicted outcome for the same individual under the control condition t = 0.

ITE_i is the Individual Treatment Effect, the estimated difference in outcome that treatment t would produce for person i, relative to no treatment. It is a counterfactual comparison made possible by the digital twin.

Allowing:

- Personalization of treatment strategies.
- Scenario testing: 'what if we had applied treatment A instead of B?'
- Estimation of population-level or subgroup-level effects.

Integration with reinforcment learning

Causal digital twins can be extended into reinforcement learning frameworks. Here, the twin simulates dynamic outcomes across time under sequences of actions. This leads to:

- Personalized treatment policy optimization,
- Value-based decision support,
- Simulation of long-term interventions (e.g., chronic disease management).

The value function is:

$$V^{\pi}(X) = \mathbb{E}\left[\sum_{t=1}^{T} \gamma^{t} Y_t \mid \pi, \mathrm{X}\right]$$

Where:
$V^{\pi}(X)$ is the value function, representing the expected sum of rewards (or clinical outcomes) over time when following policy π, starting from covariates X.
$\mathbb{E}[.]$ Is the expectation operator, indicating an average over all possible future scenarios or stochastic elements in the system.
$\sum_{t=1}^{T}$. is the summation over time steps t from 1 to T, which is the time horizon (e.g., duration of treatment, number of clinical visits).

γ^{t} is the discount factor (where $0 < \gamma \leq 1$), used to reduce the weight of outcomes that occur further in the future. It reflects preference for short-term vs. long-term outcomes.
Y_t is the reward or outcome observed at time t, such as symptom reduction, biomarker improvement, or quality-of-life score.
π is the treatment policy, which maps a patient's history (or current state) to a recommended action or treatment.
X is the initial state or covariate vector of the individual (e.g., age, diagnosis, baseline severity).

Methodology

This section outlines a structured approach for building, training, and validating digital twins as causal digital twins, especially in the context of healthcare applications. The methodology spans data preparation, model development, evaluation, deployment, and ethical oversight.

Data sources and requirements

High-quality data is foundational for digital twins. Each sample in the dataset should ideally include:

- Covariates X: patient features like age, sex, biomarkers, socioeconomic status;
- Treatment T: binary or multi-valued variable indicating the intervention (e.g., drug A vs. placebo);
- Outcome Y : clinical response, survival time, adverse event, etc.;

– Temporal dimension: for dynamic twins, data should capture longitudinal patterns.

Digital twin performance critically depends on the quality and representativeness of the data. Biases in data collection (e.g., missingness, selection bias) must be identified and mitigated.

Data preprocessing and representation learning

Data preprocessing transforms raw data into structured inputs suitable for causal modeling. Key steps include:

- Missing data imputation: e.g., using mean imputation, multiple imputation, or model-based techniques;
- Normalization and encoding: scaling continuous variables, one-hot encoding categorical variables, or embedding them into low-dimensional spaces;
- Dimensionality reduction: applying methods like principal component analysis (PCA), t-SNE, or variational autoencoders (VAEs) to compress complex covariate structures;
- Balancing covariates: via matching (e.g., propensity score matching), reweighting (e.g., IPTW), or learning balanced representations (e.g., adversarial domain adaptation).

Representation learning ensures that models retain relevant causal information while simplifying high-dimensional input spaces.

Model development and learning objectives

Digital twins are typically trained to estimate both factual and counterfactual outcomes. The predictive model $\widehat{f}$ is optimized to minimize the discrepancy between predicted and observed outcomes, as a factual loss function used during the training of a digital twin model:

$$\mathcal{L}_{factual} = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \widehat{f}(X_i, T_i)\right)^2$$

$\mathcal{L}_{factual}$ is the average loss over all observations; specifically, it's the mean squared error between observed and predicted outcomes.

Y_i is he observed outcome for individual i.

$\widehat{f}(X_i, T_i)$ is the predicted outcome for individual I, given their covariates X_i and the treatment they atually received T_i.

n is the total number of individuals in the traing dataset.

This loss function penalizes the model when its predicted outcomes deviate from actual observed outcomes, and it's the standard objective in supervised learning for regression problems. In causal inference, minimizing this loss helps the model accurately learn factual outcomes, which is crucial before trying to estimate counterfactuals.

To address the unobserved nature of counterfactuals, the following strategies are often used:

– Inverse probability weighting (IPW): reweights samples to adjust for confounding;
– Doubly robust estimation: combines IPW and outcome modeling for improved robustness;
– Adversarial training: minimizes representation bias between treated and control groups.

Common architectures include: S-learner, T-learner, X-learner, Dragonnet, GANITE etc.

Evaluation metrics

Causal digital twins must be evaluated on both predictive accuracy and causal relevance. Key metrics include:

– Root Mean Square Error (RMSE): for observed outcome prediction accuracy;
– PEHE (Precision in Estimation of Heterogeneous Effects):

$$PEHE = \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\mathcal{T}}_i - \mathcal{T}_i\right)^2$$

Where $\widehat{\mathcal{T}}_i$ is the estimated individual treatment effect (ITE) for individual *i*,
$\mathcal{T}_i$ is the true (ground-truth) individual treatment effect for individual *i*,
and n, the total number of individuals in the evaluation dataset.

PEHE measures how accurately a model predicts personalized treatment effects. A lower PEHE indicates that the model's estimated treatment effects closely match the true values, suggesting better precision in individualized causal inference:

– ATE Error: difference between estimated and true average treatment effect;
– Calibration: goodness-of-fit of predicted probabilities or risk estimates;
– Counterfactual consistency: degree of stability in predictions when inputs are perturbed slightly.

Benchmarking and validation

Model benchmarking involves assessing performance against datasets with known causal structure. Key resources include:

- IHDP: simulated outcomes from a real infant health dataset;
- Twins: counterfactual outcomes approximated via sibling comparisons;
- Jobs/LaLonde: datasets on employment programs with experimental and observational arms.

Validation techniques include:

- Cross-validation: stratified or grouped by treatment assignment;
- Simulation studies: create synthetic populations with ground-truth effects;
- Sensitivity analysis: test model robustness to violations of assumptions like ignorability.

Deployment in healthcare settings

For practical utility, digital twins must integrate into real-world clinical workflows:

- Interoperability: support for standards such as HL7 FHIR or OMOP-CDM for data exchange;
- Explainability: use interpretability tools like SHAP, LIME, ICE plots to aid decision-making;
- Real-time inference: models should offer low-latency predictions to inform clinical action;
- Continual learning: systems should adapt as new data becomes available, ensuring model relevance.

Ethical oversight and fairness

Ethical deployment of digital twins in healthcare requires rigorous governance:

- Bias audits: systematically evaluate performance across demographics (e.g., sex, race, age);
- Transparency: clearly document data sources, model assumptions, and design choices;
- Accountability: ensure human oversight over automated recommendations, especially in critical care settings.

Developing and deploying causal digital twins in healthcare raises significant ethical considerations that must be addressed systematically within the methodology.

First, transparency and explainability are essential to ensure that clinicians and patients understand the basis of treatment recommendations. Models must integrate interpretability tools, such as SHAP or ICE plots, to make the causal mechanisms and individualized treatment effects comprehensible to non-technical users.

Second, fairness and bias auditing must be an integral part of model validation. Digital twins should be evaluated across demographic subgroups (e.g., age, sex, ethnicity, socioeconomic status) to detect and mitigate disparate impacts. Strategies such as subgroup-specific performance metrics, fairness-constrained optimization, and bias mitigation techniques should be employed to ensure equitable model behavior.

Third, privacy and data protection are paramount. Given that digital twins simulate sensitive health trajectories, robust measures must be implemented to safeguard patient data. Approaches such as data de-identification, federated learning, and differential privacy mechanisms can be used to minimize the risks associated with centralized data storage and model training.

Then, the deployment of causal digital twins must uphold the principle of human oversight. Clinical decision-making should not be fully delegated to automated systems; rather, digital twins should serve as advisory tools that augment, but do not replace, physician judgment. Clear accountability frameworks must be established to delineate the roles and responsibilities of model developers, clinical users, and institutional governance bodies.

Evaluation

This section details the evaluation strategies used to assess the performance and reliability of digital twins designed as causal digital twins. A robust evaluation framework is essential to ensure that digital twins provide accurate, generalizable, and clinically useful insights.

Evaluation objectives

The evaluation process serves several purposes:

- To quantify the accuracy of outcome predictions (factual outcomes).
- To measure the reliability of estimated treatment effects (causal performance).
- To assess fairness, robustness, and consistency under varied conditions.

Predictive performance

Root Mean Square Error (RMSE).

RMSE assesses how close predicted outcomes are to observed (factual) outcomes:

$$RMSE : \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2}$$

Where.
Y_i is the observed outcome for individual i.
$\widehat{Y}_i$ is the predicted outcome from the digital twin.
And n, the total number of individuals.
Lower RMSE indicates better predictive accuracy of the factual model.

Causal performance

PEHE evaluates the precision of individualized treatment effect estimation:

$$PEHE = \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\mathcal{T}}_i - \mathcal{T}_i\right)^2$$

Where
$\widehat{\mathcal{T}}_i = \widehat{Y}_i(1) - \widehat{Y}_i(0)$ is the estimated individual treatment effect of individual i.

$\mathcal{T}_i = Y_i(1) - Y_i(0)$ is the true treatment effect (only known in simulated datasets).
And n, the number of individuals.

Lower RMSE indicates better predictive accuracy of the factual model.

ATE error compares the estimated and true average treatment effects:

$$ATE\ Error = \left| \frac{1}{n}\sum_{i=1}^{n}\widehat{\mathcal{T}}_i - \frac{1}{n}\sum_{i=1}^{n}\mathcal{T}_i \right|$$

This measures how close the model's average treatment effect estimate is to the ground truth.

Calibration and counterfactual robustness

Calibration evaluates how well predicted probabilities or continuous outcomes match the actual outcomes. This is particularly important in probabilistic models (e.g., predicting risk of adverse outcomes).
Tools include:

- Calibration curves: compare predicted risk with observed outcomes across bins.
- Brier score: a proper scoring rule for probability forecasts.

Counterfactual consistency: This concept evaluates the model's stability when small perturbations are applied to the input data.

For example, the model should not radically change its treatment effect predictions if a non-influential covariate is slightly altered. Techniques include:

- Local perturbation analysis.
- Sensitivity metrics.

Subgroup analysis

Subgroup analyses assess whether the model performs equitably across key demographic or clinical subpopulations (e.g., age groups, sex, comorbidities). Metrics like RMSE, PEHE, and ATE error can be recomputed within each subgroup to evaluate disparities in model performance.

Fairness and bias auditing

Evaluation includes testing for systematic bias in treatment recommendations or outcome predictions. Fairness audits may compare:

- Disparate impact or error rates across groups.
- Calibration within each subgroup.
- Consistency of treatment effects across protected attributes.

Fair models should not amplify existing inequities or underperform for marginalized populations.

Use of benchmark datasets

Evaluation is often performed on benchmark datasets that include partially known counterfactuals:

- IHDP: real covariates, simulated outcomes.
- Twins: matched outcomes using real-world twin data.
- Jobs: experimental treatment and employment outcomes.

Simulated datasets allow estimation of true PEHE and ATE error. Real-world datasets test robustness and external validity.

Synthesize

Having outlined the key theoretical frameworks and methodological advances underpinning causal digital twins, we now synthesize these concepts into a unified formalism that captures the essential structure of individualized counterfactual modeling and decision optimization.

The convergence of causal inference, individualized simulation, and adaptive decision-making heralds a paradigm shift in precision medicine. By embedding SCMs and the Potential Outcomes Framework within dynamic digital twin architectures, and extending these models with reinforcement learning for policy optimization, we create computational systems capable of both simulating and prescribing personalized health trajectories. This synthesis bridges prediction and intervention, allowing digital twins to not only foresee clinical futures but to shape them through counterfactual reasoning and optimal action selection.

The core principles of this causal digital twin framework can be formalized as follows:

$$\mathcal{T}_i(X_i) = \left\{\widehat{Y}_i(t) = \mathbb{E}[Y_i(t) \mid X_i] \forall t \in \mathcal{T}\right\} \Rightarrow ITE_i$$
$$= \widehat{Y}_i(1) - \widehat{Y}_i(0) \Rightarrow \pi^*(X_i)$$
$$= arg \max_{\pi} \mathbb{E}\left[\sum_{k=0}^{T} \gamma^k r(X_{i,k}, \pi(X_{i,k}))\right]$$

Where:
$\mathcal{T}_i(X_i)$ is the causal digital twin of individual i, a mapping from covariates X_i to potential outcomes.
$\widehat{Y}_i(t)$ is the predicted potential outcome if treatment ttt were assigned, for any ttt in the set of possible treatments $\mathcal{T}$.

ITE_i is the Individual Treatment Effect (difference between treated and untreated predicted outcomes).
$\pi^*(X_i)$ is the optimal personalized treatment policy based on maximizing expected cumulative reward.
$\mathbb{E}$ is the expectation operator (over stochastic elements).
$r(X_{i,k}, \pi(X_{i,k}))$ is the reward at time k given covariates and treatment decision.
γ is the discount factor ($0 < \gamma \leq 1$) valuing present rewards over future ones.
And T, the horizon (number of decision points over time).

Discussion

The development of digital twins as causal digital twins marks a transformative step in personalized medicine and data-driven healthcare [5, 16, 17]. Unlike traditional predictive models, causal digital twins not only forecast likely outcomes but also simulate alternative scenarios, allowing clinicians to answer "*what if*" questions with greater confidence [1]. One of the principal strengths of causal digital twins lies in their ability to estimate individualized treatment effects [18]. This capacity enables tailored medical decisions that go beyond average treatment effects by accounting for the heterogeneity in patient characteristics and responses (Table 2) [19].

By incorporating formal causal frameworks such as Structural Causal Models and the Potential Outcomes Framework [8, 20], digital twins can overcome common biases in observational data and provide plausible simulations of counterfactual outcomes [14]. Furthermore, digital twins integrate seamlessly with modern machine learning architectures, ranging from neural networks to tree-based methods [7], which allows for their scalable deployment across large and complex clinical datasets. Their compatibility with interpretability tools, such as SHAP values or ICE plots [21, 22], supports transparency and builds clinician trust in decision-support systems.

Despite these advantages, several challenges remain. The reliability of a digital twin model depends heavily on the quality, completeness, and granularity of the data it uses [23]. Missing data, measurement error, and unmeasured confounding can significantly compromise model performance [24]. Moreover, because counterfactual outcomes are inherently unobservable, validating these predictions remains methodologically difficult [13]. Although synthetic benchmarks and simulated data provide partial solutions [25, 26], they cannot fully substitute real-world evidence. Another challenge is the computational complexity involved in training robust models, particularly those that rely on generative or adversarial architectures [27]. Furthermore, models trained in one healthcare setting or population may not generalize well to others without further calibration or domain adaptation [28]. Finally, deploying digital twins in clinical environments raises important regulatory, legal, and ethical questions [29, 30]. These include concerns about model transparency, fairness across patient populations, and the degree of human oversight required in high-stakes decision-making [31].

Looking forward, several promising directions are emerging. The integration of digital twins into real-time clinical decision support systems could bring simulated insights directly to the point of care [32]. The development of longitudinal and dynamic twins, capable of updating as patient trajectories evolve, may further

Table 2 Illustration of the key differences between predictive and causal digital twins, highlighting the conceptual advances that underlie this shift

Aspect	Predictive digital twin	Causal digital twin
Core question	What is likely to happen next?	What would happen if we intervened differently?
Main goal	Forecast future states or risks	Estimate causal effects of interventions
Data dependency	Relies on observed associations	Requires causal assumptions and adjustment
Theoretical foundation	Statistical/machine learning models (supervised learning)	Structural causal models, potential outcomes
Output type	Predicted future outcome (factual)	Simulated outcome under alternative treatment (counterfactual)
Clinical use	Risk stratification, early warning systems	Treatment choice, policy simulation, individualized effect estimation
Validation challenge	Prediction accuracy (e.g., RMSE)	Counterfactual validity, identifiability
Decision support	Suggests likely outcome under current treatment	Recommends optimal action based on ITE or CATE

enhance the timeliness and relevance of recommendations. Hybrid models that combine expert clinical knowledge with data-driven inference are likely to improve causal validity while maintaining interpretability [33]. Digital twins could also be used to simulate public health interventions at scale, helping policymakers assess the potential outcomes of healthcare reforms before implementation. Moreover, the use of federated learning and privacy-preserving computation will be key to enabling secure model training across institutions, ensuring that sensitive health data remains protected (Fig. 2) [34].

In parallel, interpretability is a critical component in the deployment of causal digital twins, particularly in clinical contexts where decision support must be transparent and actionable [35, 36]. While digital twins can emulate both factual and counterfactual outcomes, understanding why a model recommends a specific treatment or predicts a certain response is essential for clinical trust and regulatory acceptance [29].

Tools such as SHAP (SHapley Additive exPlanations) and ICE (Individual Conditional Expectation) plots can be integrated into causal digital twins to reveal the contribution of individual covariates to predicted treatment effects [21, 22]. SHAP values, derived from cooperative game theory, provide a local explanation by quantifying the marginal impact of each feature on the model's output for a specific individual [37]. In contrast, ICE plots show how the predicted outcome changes as a single feature varies, holding others constant [38].

Beyond transparency, interpretability also facilitates model validation, error analysis, and bias auditing [39]. By visualizing and interpreting treatment effect heterogeneity, digital twins can highlight outliers, identify sources of confounding, and support counterfactual reasoning in complex clinical scenarios. Thus, integrating explainability methods not only enhances clinical usability but also reinforces the scientific credibility of causal digital twin models [31, 40].

Methodological limitations and causal assumptions

The validity of counterfactual predictions generated by causal digital twins relies on a set of causal assumptions that warrant careful consideration. Chief among these is the assumption of no unmeasured confounding, which posits that all variables jointly influencing treatment assignment and outcomes are observed and appropriately modeled [8, 15]. In real-world clinical data, this assumption is inherently strong and often unverifiable, as latent factors such as clinician judgment, patient preferences,

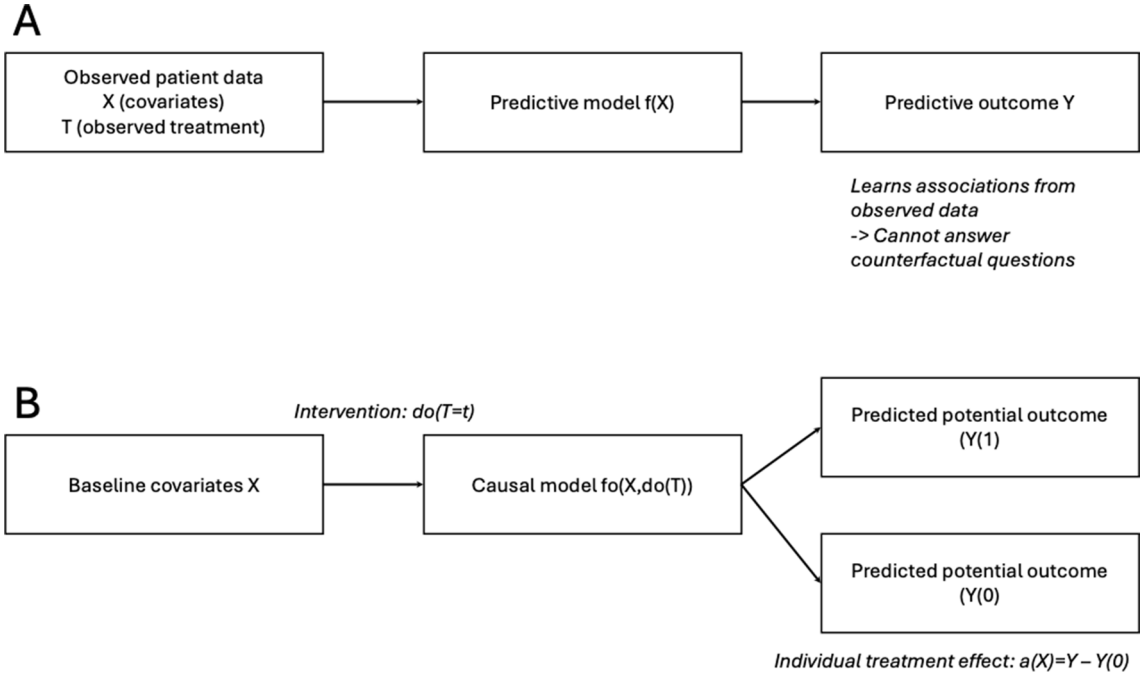

Fig. 2 Predictive versus causal digital twins: association-based prediction versus intervention-based counterfactual reasoning. **Panel A** illustrates a traditional predictive digital twin, which estimates outcomes based on observed statistical associations between baseline covariates X, observed treatment assignment T, and outcome Y. Such models learn the conditional distribution P(Y | X, T) and can predict likely outcomes under observed treatment patterns, but they do not model interventions explicitly and therefore cannot reliably answer counterfactual questions. **Panel B** depicts a causal digital twin, which represents treatment as an explicit intervention using the do-operator do(T = t) and estimates potential outcomes Y(t) under alternative treatment scenarios t ∈ {0,1}. The individual treatment effect (ITE) is defined as τ(X) = Y(1) − Y(0), and the conditional average treatment effect (CATE) as E[Y(1) − Y(0) | X]. By distinguishing association from intervention, the causal digital twin enables counterfactual reasoning and individualized, causally interpretable clinical decision support

or unrecorded disease severity may influence both treatment decisions and outcomes [15, 41].

Violations of this assumption may lead to biased estimates of treatment effects, potentially compromising the reliability of individualized recommendations [42, 43]. As a result, causal digital twins should not be interpreted as definitive decision-makers, but rather as decision-support tools whose outputs must be contextualized within clinical expertise.

Additional assumptions, including positivity (sufficient overlap between treatment groups), consistency, and correct model specification, further constrain applicability. Limited overlap may restrict inference to specific patient subgroups, while model misspecification can distort both factual and counterfactual predictions.

To mitigate these limitations, several strategies can be employed, including sensitivity analyses to assess robustness to unmeasured confounding, incorporation of domain knowledge in causal graph specification, triangulation across multiple data sources, and prospective validation where feasible. Ultimately, maintaining human oversight and emphasizing interpretability are essential to ensure that causal digital twins enhance, rather than replace, clinical judgment [23].

Causal assumptions, failure modes, and mitigation strategies

Individual-level counterfactual inference using digital twins rests on a set of structural assumptions that are both necessary for identification and fragile in realistic clinical environments. Making these assumptions explicit is essential for interpreting estimated individual and conditional average treatment effects as causal quantities rather than purely predictive constructs.

The assumption of consistency requires that the potential outcome under a given intervention corresponds to the outcome observed when that intervention is implemented in practice. In clinical settings, however, interventions are rarely atomic: treatment effects may vary with dose, timing, adherence, concomitant therapies, and provider behavior. Violations of consistency induce ambiguity in the definition of the causal estimand itself, rendering individualized effect estimates ill-posed. A principled mitigation strategy consists in formulating interventions as well-defined dynamic treatment regimes and modeling treatment intensity and adherence explicitly within the state transition equations of the twin.

Exchangeability (or ignorability) conditional on observed covariates is required for causal identification [8, 44]. This assumption is particularly vulnerable in healthcare, where latent constructs such as frailty, clinician judgment, subclinical disease severity, and socioeconomic context often influence both treatment assignment and outcomes. When violated, estimated ITEs and CATEs may exhibit systematic bias that is not detectable through conventional predictive validation. Beyond high-dimensional adjustment, methodological safeguards include the incorporation of negative and positive control outcomes or exposures, formal sensitivity analyses to quantify the magnitude of unmeasured confounding required to overturn conclusions, and partial identification approaches that yield bounded rather than point estimates [45].

The positivity assumption requires sufficient overlap in treatment assignment across the covariate space. In practice, clinical contraindications, institutional protocols, and strong physician preferences may create structural violations whereby certain patient profiles are deterministically assigned to a single treatment. In such regions, individualized counterfactual predictions become extrapolations unsupported by empirical evidence. Diagnostic evaluation of overlap, restriction of inference to regions of common support, and explicit reporting of effective sample size are therefore necessary to prevent overconfident decision support.

Digital twins implicitly assume correct model specification at multiple levels: statistical risk models, mechanistic submodels, and their coupling through state-space representations. Misspecification may arise from incomplete biological knowledge, dataset shift, or simplifications introduced for computational tractability. Importantly, low predictive error on observed outcomes does not guarantee valid counterfactual reasoning. Model criticism should therefore extend beyond discrimination metrics to include calibration of predicted trajectories, posterior predictive checks, invariance testing across environments, and stress testing under controlled perturbations of physiological parameters [46].

Causal digital twins presuppose a degree of transportability and temporal stability of causal mechanisms [47]. Shifts in population case-mix, standards of care, diagnostic practices, or healthcare infrastructure may invalidate previously learned relationships. Formal transportability analysis, reweighting schemes, domain adaptation techniques, and continual learning frameworks are thus required to maintain causal interpretability under deployment drift. Collectively, these considerations imply that causal digital twins should be conceptualized as epistemic instruments whose outputs are contingent on unverifiable assumptions and model abstractions [48]. Their safe clinical integration therefore depends not only on algorithmic performance, but on systematic assumption auditing, uncertainty propagation, and transparent communication of the limits of causal interpretability (Table 3).

Table 3 Each assumption may be violated in realistic clinical deployments

Assumption	Failure mode	Clinical consequence
Exchangeability	Hidden confounders (frailty, clinician preference, undocumented severity)	Biased ITE/CATE estimates
Positivity	Rare treatments or exclusion criteria	Extrapolation beyond data
Consistency	Treatment heterogeneity (dose, adherence, co-interventions)	Misaligned causal estimand
Model specification	Oversimplified physiology, dataset shift	Systematic prediction error
Transportability	Different hospital practices, populations	Loss of external validity

Conclusion

Causal digital twins represent a pivotal advancement in precision medicine, enabling not only the prediction of future health trajectories but also the simulation of individualized counterfactual scenarios. By integrating structural causal models, potential outcomes frameworks, and reinforcement learning into dynamic digital architectures, this work offers a unified foundation for optimizing personalized treatment decisions. While challenges related to data quality, counterfactual validation, and fairness remain, the methodological synthesis proposed here moves digital twins beyond prediction toward actionable and adaptive decision support systems, paving the way for the next generation of AI-driven healthcare. Moreover, while causal digital twins offer powerful counterfactual insights, their clinical impact ultimately depends on careful validation of causal assumptions and responsible integration into human-centered decision-making workflows.

Abbreviations

ITE Individual Treatment Effect
CATE Conditional Average Treatment Effect
ATE Average Treatment Effect
DAG Directed Acyclic Graph
SCM Structural Causal Model

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